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Military expenditure and economic growth: A meta-analysis

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ABSTRACT

Meta analysis is conducted to review 32 empirical studies with 169 estimates of the effect of military expenditure on economic growth. We formulate four hypotheses to examine the empirical evidence and to provide overall conclusions while controlling for systematic heterogeneity in the studies reviewed. The hypotheses are: (H1) Military expenditure reduces economic growth; (H2) Military expenditure is detrimental to economic growth in less developed countries (LDCs); (H3) The effect of military expenditure on economic growth is positive and (H4) The effect of military expenditure on economic growth is non-linear. We find that the hypothesis of a negative military expenditure–growth relationship is not supported for both LDCs and in general, while a positive effect of military expenditure on economic growth is supported for developed countries. The hypothesis of a non-linear military expenditure–growth relationship is confirmed. The main sources of study-to-study variation in the findings of military expenditure and economic growth literature are attributable to the sample, time periods, and functional forms.

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1. Introduction

The effect of military expenditure on the economy is a controversial area of research among economists. There are several channels defined through which military expenditure may affect economic growth. Conceptually, each channel may lead to different conclusions and thus the net effect is ambiguous. The issue then is an empirical one. However, the empirical findings have not produced a conclusive result either: their conclusions are that the effect may be negative or positive or insignificant.

Since late 1970s there has been a substantial research in this area. Arguably, the general view has been that even if there are positive effects, these effects would be offset by negative ones. Broadly defining, on the one hand there are security-related positive effects on economic factors and supply-side spillovers; on the other hand there is the negative effect of diverting resources away from the civilian economy.

This paper is the first to provide a substantial quantitative survey of military expenditure and economic growth literature by conducting a meta regression analysis. In a closely related study, Nijkamp and Poot (2004) apply a different meta-analysis technique (i.e. rough set analysis) to examine the effect of disaggregated government expenditure on economic growth, which also includes military expenditure. They look at the studies that are published between 1983 and 1998 in refereed journals. However, our study focuses only the military expenditure and economic growth literature to address between study variations and the possible causes of systematic heterogeneity.

Deger and Sen (1995), Ram (1995) and Dunne (1996) offer a detailed survey of the military expenditure–economic growth literature and conclude that the effect of military expenditure on economic growth varies depending on the design of empirical strategy. Nevertheless, these reviews are not able to statistically identify what type of research designs lead to systematic

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heterogeneity among reported findings. For instance, Ram (1995) suggests that there may be some structural heterogeneity across the 1960s, the 1970s, and the 1980s, therefore using meta-analysis we can test if this is the case. Meta-analysis is a statistical tool that allows for a systematic analysis of diverse empirical findings, while accounting for differences among reported results of individual studies.

The following are the important hypotheses that have been developed and tested in the literature:

- H1: Military expenditure reduces economic growth — defence burden.
- H2: Military expenditure is detrimental to economic growth in less developed countries.
- H3: The effect of military expenditure on economic growth is positive.
- H4: The effect of military expenditure on economic growth is non-linear.

The first hypothesis is about crowding out effects. The literature suggests that there is a trade-off between productive (e.g. investment and education) and unproductive (defence outlays) government spending. The second implies that there are regional and country differences that play an important role in determining the effect of military expenditure on economic growth in least developed and developing countries. The third hypothesis is about supply-side spillovers and aggregate demand effect. The final one is due to the first three hypotheses. The existing empirical literature has extensively tested each of these hypotheses but it has not been possible to accept or reject any of these hypotheses. In this paper, a meta-analysis technique is conducted to find an overall, statistically valid conclusion among the reported results of the military expenditure–economic growth literature.

Thus, there are two important issues which are addressed in this survey: first, whether there exists a 'genuine' relationship (to be defined) between military expenditure (milexp henceforth) and economic growth. Second, to examine the sources of variations in the milexp-growth literature. One can define the sources of variations into three sub categories; firstly, the choice of theoretical models, which provide the basis of the empirical channel to be tested i.e. demand–supply side, supply side, ad hoc and so forth; secondly, the design of the empirical study, i.e. sample, time period and estimation technique; and finally the other control variables included. Thus, the aim is to account for all these study-to-study differences and quantify the net overall effect.

The paper is organised as follows. Section 2 provides an overview of conceptual considerations with the empirical specifications. This follows with a discussion of the primary studies included in the meta-regression in Section 3. Section 4 summarizes the meta-analysis methodology and uses fixed and random effects and meta-regression analysis to assess the military expenditure–growth literature. Finally, Section 5 presents some conclusions and suggestions for future research.

2. Main conceptual and econometric approaches

Since 1990s, there is much discussion about the composition of government expenditure and which components of government expenditure are growth-enhancing. In effect, the impact of disaggregated government spending on growth is commonly studied, which includes milexp, see Aschauer (1989), Easterly and Rebelo (1993), Devarajan et al. (1996), Mulas-Granados et al. (2002), Bose et al. (2007). However, each segment of government spending may have varying effects on the long-run economic growth. Arguably the emerging view from these studies is that education, infrastructure and capital expenditure are among those with positive effect. These are commonly referred as the *productive* component of government expenditure. By and large, theoretically and empirically, the effect of milexp is expected to be negative.

Nevertheless, as already noted, the economic effects of milexp are too complex to draw a general consensus; a *priori*, the effect can go either way. There are many factors contributing to this such as the theoretical frameworks underlying the empirical studies. To obtain insight, three commonly conducted empirical specifications together with the theoretical frameworks are briefly reviewed in this section: namely the neo-classical supply-side model of Feder–Ram; the Keynesian demand and supply model; and Barro-type growth regressions. We discuss these in turn.

2.1. The three approaches

Feder (1983) develops a model to evaluate effects of import and export sectors' on growth. This model is extended by Ram (1986) and Biswas and Ram (1986) to look at defence and non-defence sectors and their impact on growth. This approach is developed to investigate the claims related to the positive externality effect of milexp on growth.[1]

The model assumes that there are two sectors in the economy, civilian (C) and defence (M). The inputs in both industry are labour (L) and capital (K) which are allocated to each sector. The production functions for each sector are:

$$M = M(L_M, K_M) \qquad C = C(L_C, K_C, M)$$

[1] Although this is one of the main approaches used in this literature, the primary studies that apply this method are not included in the quantitative analysis due to the reasons discussed later in the paper.

The output of the economy is $Y = M + C$. The model also assumes that the marginal productivities of capital and labour in each sector might differ:

$$\frac{M_K}{C_K} = \frac{M_L}{C_L} = 1 + \delta$$

In essence, $\delta > 0$ implies that capital and labour inputs are more productive in the defence sector. Using this model, there are two issues that could be tested, namely a positive externality effect and question of whether resources are more productive in the defence sector or not. Two empirical specifications are used in the empirical literature of Feder–Ram model, one of which is:

$$\dot{Y} = \alpha \frac{I}{Y} + \beta \dot{L} + \left(\frac{\delta}{1+\delta} - \theta \right) \frac{M}{Y} \dot{M} + \theta \dot{M}$$

where a variable with dot notation is a rate of growth of the variable. $\theta = C_M \left(\frac{M}{Y-M} \right)$ is the elasticity of civilian output, (C), with respect to military output, (M), and C_M, the marginal product of military output, indicates the externality effect. Therefore, if C_M and δ are positive, the effect of milexp on growth is expected to be positive. This model has been used by many; Alexander (1990), Atesoglu and Mueller (1990), Huang and Mintz (1991), Ward and Davis (1992), Cuaresma and Reitschuler (2004), among others. However Dunne et al. (2005) suggest that there are important shortcomings of the model in terms of the interpretation of results as well as econometric techniques.[2]

Although, there may be positive direct effects on the economy, either as a result of the positive spillovers or through a Keynesian aggregate demand model (Deger, 1986), there is the opportunity cost of milexp or the crowding-out effect. This argument is broadly associated with the limited resources being allocated to a sector that creates forgone opportunities in the other sector(s). Therefore, this effect is seen as the indirect negative effect of milexp on other government expenditure and the main argument evolves around the negative effect on total investment. Deger and Smith (1983) and Deger (1986) suggest a simultaneous equations model (SEM) that considers both supply and demand effects of milexp and allows for the assessment of direct and indirect effects of milexp on the civilian economy. In other words, due to the interrelation between growth, milexp and investment, the application of a single-equation methodology is limited since it does not allow for quantifying the total effect of milexp on growth. A general model given here is a four-equations SEM:

$$\begin{aligned}
g_{it} &= a_0 + a_1 s_{it} + a_2 m_{it} + a_3 B_{it} + \sum_j \alpha_j^g x_{j,it}^g + u_{it}^g \\
s_{it} &= b_0 + b_1 m_{it} + b_2 g_{it} + b_3 B_{it} + \sum_k \alpha_k^s x_{k,it}^s + u_{it}^s \\
B_{it} &= c_0 + c_1 m_{it} + c_2 g_{it} + \sum_l \alpha_l^B x_{l,it}^B + u_{it}^B \\
m_{it} &= d_0 + \sum_n \alpha_n^m x_{n,it}^m + u_{it}^m
\end{aligned}$$

where g is the growth rate of income per capita, s is the saving ratio, m is the share of milexp in GDP, B is the trade balance, x denotes a set of predetermined variables in each equation, and u is the error term, (Deger and Sen, 1995). Based on this analysis the positive effect is through the coefficient, a_2 and then the milexp variable in the second and third equations would allow to assess the indirect effect, which is expected to be negative. Deger (1986) argues that an increase in milexp might result in higher inflation or an increase in taxation and this in effect may result negative effect on investment. The negative effect on investment has knock-on effect on growth. There are a number of studies that applied SEM; see for instance Deger and Sen (1983), Deger (1986), Lebovic and Ishaq (1987), Gyimah-Brempong (1989), Dunne and Mohammed (1995), Galvin (2003).[3]

Finally, considerable empirical studies of milexp-growth apply some variant of Barro-type growth specifications.[4]

$$g_{it} = a_0 + a_1 m_{it} + \sum_j \alpha_j x_{j,it} + \sum_k \alpha_k z_{k,it} + u_{it}$$

where g is the growth rate of income per capita and m is the ratio of milexp to GDP. x_j is a set of common variables used in the economic growth literature and has theoretical foundations: these are investment, initial level of income, population growth and education. z_k is a set of variables of interest in milexp-growth literature such as square of milexp, dummy for war, threat and so on. This specification is commonly used in general growth literature. This reformulation of the Barro model is used by Brumm (1997), Stroup and Heckelman (2001), Aizenman and Glick (2006), Yakovlev (2007), among others.

[2] For a thorough discussion on Feder–Ram model see Dunne et al. (2005).

[3] There are also numerous studies that focus on a potential feedback effect from growth to milexp using atheoretical models, such as Granger-causality and more recently vector autoregression models. Unlike the simultaneous-equation models, they do not impose *a priori* any causal relationship among milexp and growth. See for instance Joerding (1986), Chowdhury (1991), Dakurah et al. (2001), Abu-Bader and Abu-Qarn (2003), Dunne and Nikolaidou (2005), among others. For detailed discussion see Dunne and Smith (2010).

[4] Some also apply the augmented Solow growth model, such as Knight et al. (1996), and Yakovlev (2007).

2.2. Criteria for selecting the empirical studies for the meta-analysis

In selecting the studies to be included in the meta-analysis of milexp and growth literature we use the following criteria: (*i*) The studies considered in the meta-analysis are those which look at the direct effect of milexp on economic growth; in which milexp is measured as the percentage of GDP and in the regression analysis, the dependent variable is economic growth and the independent variable is milexp. Thus, studies that use the growth rate of milexp as a regressor are left out, i.e. Feder–Ram[5] and Granger causality based studies. Furthermore, as discussed above, studies may have taken different channels to analyse direct and indirect effects of milexp on the economy. Although, the studies that apply a simultaneous equations model are included in the analysis, it is not possible to consider the net effect from these studies.[6] As a result, the direct effect is used in the analysis. (*ii*) Both published or unpublished studies are used in this analysis. Arguably, it is expected that there are relatively small errors in peer-reviewed studies, and therefore these studies are somehow preferred in the meta-analysis. However, this selection criteria is generally criticised, since there may be a tendency towards significant studies only to be published.[7] (*iii*) Finally, since the empirical studies are only included and the *effect size* statistic is the partial correlation coefficient, each reported empirical result should also provide information such as the t-statistic or standard error and degrees of freedom or sample size. This will allow to calculate a correlation coefficient for individual estimates. As result, the studies that only report the estimated coefficient are left out, even though they satisfy (*i*) and (*ii*) above.

Having looked at the commonly used econometric specifications and defined the criteria for selecting studies for the analysis, the next section provides a discussion of empirical studies that are used in the meta-analysis. In the Appendix A, a list of studies used in the meta-analysis is given in Table A.1 with underlying conceptual models, data types and periods.

3. Review of the studies included in the meta analysis

The first empirical milexp growth study is Benoit (1973, 1978). Benoit (1978) studies 44 less-developed countries (LDCs) to look at the milexp and growth relationship empirically and he finds a positive link. Benoit (1978) proposes a neo-classical supply side explanations: milexp can affect growth in two directions; negatively by taking away the resources which may be better used in civilian economy and positively by providing jobs and increasing employment, involving in infrastructure, training and research and development, (R&D). In the case of LDCs the benefits of milexp offset the negative ones.

The methodology and conclusions of Benoit (1973, 1978) is criticised for being a fairly intuitive and informal, see Deger (1986). There are also some critiques about the sample of Benoit's study being biased towards LDCs. Biswas and Ram (1986), in particular, revisit Benoit's methodology and apply it to poorer LDCs. The findings show that the inverse relationship between milexp and growth in poorer LDCs is strong. However, Benoit's work induced more research and the subsequent research has been greatly influenced by the postulates of Benoit.

In a modified Harrod–Domer growth model, Lim (1983) re-examines Benoit's study. He expands the sample from 44 LDC's to 54 for the period of 1965–73 and conducts a cross-country regression method. Lim (1983) assumes that milexp and investment are negatively related and introduces foreign capital inflow (FCI), where FCI may control investment and milexp together. Thus positive relationships do not mean that milexp would not crowd out investment. Based on the results obtained from his estimations he concludes that there is a negative effect of milexp on growth.

As one of the claims of Benoit (1978) for having a positive correlation between milexp and growth is because of an increase in utilisation of unused capital,[8] Faini et al. (1984) attest this claim, whereby a capacity utilisation rate and absorptive capacity limit variables are defined. Faini et al. (1984) apply a demand side Keynesian model using national account equation. They show that milexp can influence investment negatively, hence growth of output, through absorptive capacity. They carry out a panel fixed-effect estimation method using data from 69 countries over the period of 1952–70 and the results indicate an increase in milexp lowers growth.

As noted in the preceding section, the interaction between milexp and investment can have a knock-on effect on growth. Deger (1986) and others use demand–supply models to address this issue. Deger (1986) sets out three equations with endogenous variables being growth, investment, and milexp. Using three-stage least squares (3SLS), she estimates a model for 50 LDCs for the period of 1965–73. The findings show that investment and milexp has a positive direct effect on growth, but there is a negative relationship between milexp and investment. Then, the overall net effect of milexp on growth is found to be negative.

The simultaneity/interdependence issue is one of the problems associated with a single equation cross-country model. The other weakness associated with this type of approach is the heterogeneity issue. To accommodate this, time-series studies of a single country and panel studies are commonly used. Faini et al. (1984), Chan (1988), Antonakis (1997), Heo and DeRouen (1998) and DeRouen (2000) are among those carry out a time-series analysis. Whereas, Cappelen et al. (1984), Landau (1986), Lipow and Antinori (1995), Blomberg (1996), Stroup and Heckelman (2001), Kollias et al. (2007), Yakovlev (2007), apply a cross-country time series model. Furthermore, some studies consider relatively more homogenous samples. These are studies by Cappelen et al. (1984)

[5] The Feder–Ram model based empirical studies estimate a coefficient for a composite term of growth and level of milexp. Therefore, it is not possible to calculate comparable effect size from these studies.

[6] Due to insufficient reporting of statistics it is not possible to use the net effect from these studies.

[7] This is known as the *publication bias*. A "funnel plot" is used to check for possibility of publication bias. This will be explored in detail later in the paper.

[8] The reasoning is based on aggregate demand, an increase in government expenditure via increase in milexp results use of unemployed resources, which affects growth positively.

and Landau (1996), in which both use a sample of OECD countries and Kollias et al. (2007) concentrates on EU countries. Gyimah-Brempong (1989) and Dunne and Mohammed (1995) particularly look at a sample of African countries rather than a general of sample LDCs. Gyimah-Brempong (1989) applies a cross-section time series data methodology to a sample of 39 Sub-Saharan African countries between 1973 and 1983. There is no significant direct effect but significant indirect negative effect of milexp on growth.

Due to the mixed findings, some researchers argue that there exists a non-linear relationship between milexp and growth. A number of recent empirical studies appears to support this; Landau (1993), Stroup and Heckelman (2001), Aizenman and Glick (2006), and Kalaitzidakis and Tzouvelekas (2007). Stroup and Heckelman (2001) consider a non-linear relationship between milexp and growth. They extend the Barro-type growth model to include milexp and apply panel estimations using data from 44 African and Latin American countries from 1975 to 1989. To capture the non-linearity of milexp on growth, the squared value of milexp is included in the estimation specification. The main postulate is that the positive effect of milexp on economic growth would diminish as milexp continues to increase. Furthermore the net effect of high milexp could be negative as a result of the opportunity cost of diverting resources away from the civilian sector.[9] Therefore, they use a quadratic form and include the squared value of milexp. By applying panel estimation they take into account the cross-country variations in lost human capital and public sector inefficiencies. The findings support a non-linear relation between growth and milexp — low level of milexp increases growth and high level of milexp results in a reduction in growth.

Although Stroup and Heckelman (2001) advocate the non-linear relationship between growth and milexp, the study do not account for threat; therefore it may not actually reflect those countries with a high level of threat, which in turn results in a high milexp. In a study by Aizenman and Glick (2006) however, non-linearity between growth and milexp is associated with the degree of security and this is related to the level of threat. This view raises questions of whether it is possible to determine the level of demand for milexp. The demand for milexp for each country may differ; it may go up in the presence of wars, regional conflicts, internal or external uncertainties, the possibility of future threats or it may be completely due to the political requirements of the present regime. If, arguably, the role of the military is only to safeguard the country from possible threats, then if there is no hazard either at present or in the foreseeable future then demand for milexp should be reduced and spending more on military would be a bad allocation of resources. The model specifies that if there is a threat (and resulting insecurity) above a threshold value, then a country benefits by increasing its milexp. The results show that introducing milexp into the growth equation together with the threat variable seems to justify government spending on military when there is a high level of threat. However, the threat variable fails to take into account civil and ethnic wars.[10]

In a model of endogenous growth, Kalaitzidakis and Tzouvelekas (2007) re-examine the potential non-linear effect of milexp on economic growth and empirically test the predictions of the model for 17 OECD and 38 Non-OECD countries for the period of 1980–1995. The results are presented for the whole sample of developed and developing countries and two sub-samples; i.e. OECD and Non-OECD. The findings confirm the inverse U-shaped relationship among milexp and economic growth. Blomberg (1996) also uses a model of endogenous growth model to investigate the linkages among milexp, political instability and economic growth. In the model, milexp and political instability are expected to affect economic growth negatively. In addition, a positive *spin-off* effect of milexp on economic growth is expected since milexp is seen as an insurance against political instability. In a panel of 70 countries over the period of 1967 to 1982, Blomberg (1996) shows that *crowding out* effect is greater than the *spin-off* effect and there is a weak effect of milexp on growth.

Finally, several studies analyse milexp-growth nexus together with the other government expenditure, i.e. investment, education, health and so forth. In a recent paper, Bose et al. (2007), look at the disaggregated government expenditure using panel of 30 developing countries. They define six expenditure variables; namely capital, current, education, transport and communication, defence, other expenditures. Their findings indicate a positive significant impact of milexp on growth.

4. Meta analysis

The work of Benoit had a big impact; not only was it the first empirical investigation of milexp-growth nexus, but the positive significant results were not expected. Since then, Benoit's study has been under criticism and more studies have emerged to challenge his results. Although the subsequent studies find the adverse effect of milexp on growth, this conclusion has not been statistically robust. However the tendency among researchers to look more favorably on findings with a negative, statistically significant effect of milexp on growth could lead to incorrect conclusions.

In a survey, Dunne (1996) suggests that the empirical work considering both supply and demand side effects could be considered to reach a limited consensus that the net effect is negative. Dunne (1996) also emphasizes that the results among studies may vary due to sample size, method applied, time period, other control variables and the functional form used. Therefore, the empirical studies must be interpreted with underpinning hypotheses tested and the other conditioning variables used. In narrative surveys it is not possible to quantify the differences among studies that are testing the same relationship, whereas meta-analysis allows us to consider the specifications used which, in turn, produces a relatively more rigorous overall result.

[9] The positive effect of milexp may be arising from providing security would not be sufficient to respond the opportunities foregone.

[10] Civil wars may have a negative effect on economic growth more than international wars, see Murdoch and Sandler (2002), Collier and Hoeffler (2002) and Kang and Meernik (2005).

4.1. Effect sizes analysis: fixed and random effects

To collect the relevant empirical milexp-growth studies for meta-analysis, we use various search methods. Firstly, we search for papers citing Benoit (1973) and Benoit (1978). Secondly, we check the references in review papers, among others Ram Deger and Dunne. Finally, we search the EconLit, JSTOR, Repec and Google scholar using the key words: "defence (defense) expenditure or defence (defense) spending or defence burden or military expenditure or military spending" and " economic growth or economic development".

To allow for comparisons among each independent study, an *effect size* statistic is calculated, which has no dimensionality. The *effect size* statistic used here is the *partial correlation coefficient* (PCC)[11]. There are 32 studies that are included in the meta-analysis with a total of 169 estimates, i.e. in total there are 169 partial correlations. 40% of the 169 estimates find a negative relationship with only 38% statistically significant; whereas, 60% of the 169 estimates find a positive relationship between milexp and growth, with almost half of these are statistically significant.[12]

Table 1 represents the results of the weighted mean effect sizes within individual studies and the mean ES, $\overline{ES}$, for overall 169 estimates.[13] Based on the empirical studies in Table 1 the overall ES is found to be 0.056. Using the interpretation provided by Cohen (1988),[14] although the effect is positive and statistically significant, it has no economic significance, 0.056<0.10. This estimate is valid if the assumption of homogeneity is not violated and there is no publication or selection bias.

To investigate the robustness of fixed effects results, a homogeneity test is performed. The null hypothesis is that the true ES is the same for all the milexp-growth studies in the sample, versus the alternative hypothesis that at least one of the effect sizes differs from the rest of the sample.[15] The result of Q-test is produced in Table 2.[16] The test result is highly significant implying that the assumption of a homogenous distribution of effect sizes is not valid for the milexp-growth studies.[17] In what follows, the test implies that there are variations among effect sizes that is beyond the sampling error. Hence, a random effects model must be used.[18] Table 2 shows both the fixed and random effects models for the milexp-growth studies. The mean ES for the random effect model, $\overline{ES}_{random}$, is found to be 0.066, which is slightly greater than $\overline{ES}_{fixed}$ and still very small to have any economic significance.

Stanley (2005) indicates that researchers may consider either statistically significant results or incline towards a specific direction and magnitude of an effect. Drawing upon this, the effect sizes are further investigated for the possibility of publication bias. A funnel plot is used to graphically examine this issue. The graph is a scatter plot of effects sizes against their precision or sample size. There may be no publication bias if it emerges a shape of an inverted funnel with no asymmetries. Fig. 1 is the funnel plot of the estimated effect sizes of milexp-growth studies against their precision, $\frac{1}{SE_i}$.

The vertical solid line at the center of the graph indicates the weighted mean effect size, 0.056. The visual inspection of the funnel graph does not show much sign of asymmetries with reference to the center line. This may suggest that there is no publication bias in the milexp-growth studies. However, it should be noted that at the lower level of the graph the effects sizes on the left tail are much more dispersed than the right tail. There is some clustering of effect sizes on the right hand side of the graph. This visual inspection of the funnel graph alone will not guarantee that the literature is free of publication bias. However, the initial conclusion is that publication bias might not be present in milexp-growth studies.

To explore this thoroughly, a more formal test is conducted, the funnel asymmetry test (FAT), (Egger et al., 1997; Stanley, 2005, 2008). The test is based on the *t-statistics*.

$$t_{ij} = \beta_0 + \beta_1\left(1/SE_{ij}\right) + e_{ij} \tag{1}$$

where t_{ij} is the *t*-value of the estimated coefficient from estimate *i* of study *j*. The intercept, β_0, and slope, β_1, coefficients are to be tested if they are statistically different from zero. There is publication bias, if β_0 is statistically different from zero at conventional

[11] Based on Greene (1993), if we have a regression result with t-statistics t_{x1} and the degrees of freedom *df* then the formula below is used to compute the partial correlation between *y* and x_1: $r_{yx_1} = \sqrt{\left(t_{x_1}^2\right)/\left(t_{x_1}^2 + df\right)}$ where r_{yx1} is the *ES* to be calculated from individual studies.

[12] Fig. A.1 shows the histogram of partial correlation coefficients in the sample.

[13] The mean effect size, $\overline{ES}$, is calculated using the following formula: $\overline{ES}_{fixed} = \left(\sum w_{ij}ES_{ij}\right)/\left(\sum w_{ij}\right)$ where ES_{ij} is the estimated ES_i in study *j*. $w_{ij} = 1/(SE_{ij})^2$ is the inverse variance weight for the size *ij*. This is a model of fixed effects meta-analysis due to using the inverse variance of the estimates as a weight.

[14] Cohen (1988) suggests that if the ES based on correlation coefficient the absolute value of mean effect sizes are; $r \leq 0.10$ small, $r = 0.25$ medium, and $r \geq 0.40$ large.

[15] This assumption is very strict. As it has been discussed in Section 3., the results vary enormously in empirical findings of milexp-growth, therefore this assumption is relaxed and random effects model is employed.

[16] The Q-statistic is distributed as a χ^2 with *I* - 1 degrees of freedom with *I* being the number of effect sizes: $Q = \sum w_{ij}\left(ES_{ij} - \overline{ES}_{fixed}\right)^2 \sim \chi^2_{I-1}$ ES_{ij} is individual ES for $i = 1,\cdots,I$ and $j = 1,\cdots,J$, $\overline{ES}_{fixed}$ is the weighted mean for ES_{ij}. *Q* gives the total variance and $df = I - 1$ gives the expected variance, hence the difference, $Q - I - 1$ is the excess variance. Note that, in the fixed effects model, the between study variation is 0, i.e. $Q \leq I - 1$, hence variation among study is due to within study variation.

[17] In general, it is highly likely that this assumption is violated in empirical economics. See other meta-analysis examples in empirical economics like Abreu et al. (2005), Doucouliagos and Ulubasoglu (2006), Nijkamp and Poot (2004), de Dominicis et al. (2008) and also the special issue on meta-analysis in the Journal of Economic Surveys 2005, volume 19, issue 3.

[18] In this case, each observed ES differs from the population mean by subject-level sampling error and random error term. Hence the weight in the random effects model is given by: $w^*_{ij} = 1/\left(\tau^2 + SE^2_{ij}\right)$ where τ^2 is the between study variance. The associated weighted mean and the standard error of random effects model are: $\overline{ES}_{random} = \sum\left(w^*_{ij}ES_{ij}\right)/\sum w^*_{ij}$ and $SE_{\overline{ES}_{random}}$ 13 $= \sqrt{1/\sum w^*_{ij}}$.

Table 1
Average effect within each study and overall combined fixed and random effects.

Study	No. of estimates within the study	Average effect size
Benoit (1978)	3	0.383
Deger and Smith, 1983	5	0.131
Lim (1983)	9	0.078
Deger and Sen (1983)	1	0.146
Faini et al. (1984)	1	−0.120
Cappelen et al. (1984)	8	0.079
Landau (1986)	12	0.060
Biswas and Ram (1986)	6	0.198
Deger (1986)	5	0.330
Lebovic and Ishaq (1987)	4	−0.129
Chan (1988)	3	0.089
Grobar and Porter (1989)	5	0.128
Gyimah-Brempong (1989)	2	0.038
Looney (1989)	2	−0.284
Landau (1993)	29	0.173
Dunne and Mohammed (1995)	3	0.006
Lipow and Antinori (1995)	2	0.192
Blomberg (1996)	1	−0.042
Knight et al. (1996)	4	−0.100
Landau (1996)	11	0.285
Brumm (1997)	2	0.316
Antonakis (1997)	2	−0.456
Heo and DeRouen (1998)	5	−0.101
DeRouen (2000)	1	0.168
Stroup and Heckelman (2001)	5	0.202
Galvin (2003)	9	−0.156
Aizenman and Glick (2006)	7	−0.205
Bose et al. (2007)	4	0.278
Kalaitzidakis and Tzouvelekas (2007)	3	0.095
Kollias et al. (2007)	2	0.156
Yakovlev (2007)	10	−0.148
Looney and McNab (2008)	3	−0.129
Total	169	
Fixed Effects		0.056
Random Effects		0.066

significance levels, this is known as the FAT. β_0 also gives the direction of bias.[19] Table 3 reports the results of FAT test using the weighted least squares (WLS), clustered data analysis (CDA), and hierarchical linear model (HLM).[20] CDA is used to obtain robust standard errors since there is a problem of dependency in the data. Therefore the estimated coefficients in CDA do not differ from WLS. However, the preferred estimation method is HLM since it models both within study variation and between study.

The columns 1, 2 and 3 present the results for Eq. (1) using WLS, CDA and HLM respectively. The intercept, β_0, is statistically insignificant in all models estimated. Thus, the results suggest that there is no publication bias in milexp-growth studies, which is consistent with the funnel graph 1. However this issue is further investigated using a multivariate meta-regression analysis in the subsequent section.

The issues of heterogeneity and publication bias are statistically necessary to explore and understand the variations among reported research findings. However, in the milexp-growth literature the issue of whether there is a 'genuine' relationship or not emerges to be more important for policy-makers. To test the 'genuine' empirical effect, a *precision effect test* (PET) is conducted.

The PET test is based on the slope coefficient, β_1, in Eq. (1). The test implies that if there is a 'genuine' empirical effect the slope coefficient is statistically different from zero.[21] In what follows, Table 3 also reports the results for PET. Table 3 shows that there exists a 'genuine' empirical effect but this is very small, which is the same results as in the random-effects meta analysis.

[19] Stanley (2005) argues that $1/SE_{ij}$ is an estimated value thus it contains some level of random sampling error, which produces a bias result when it is estimated using an ordinary least squares (OLS) regression. To overcome this it is suggested to use the square root of the sample size or degrees of freedom as an alternative to inverse standard errors, $1/SE_{ij}$.

[20] The high correlation between square root of sample size (N) and $1/SE_{ij}$ suggests square root of sample size is a good proxy. The calculation are run again with square root of sample size and results do not change.

[21] There is another test for testing significant empirical effect, meta-significance test (MST). The test is based on the following regression specification: $ln(|t_{ij}|) = \alpha_0 + \alpha_1 ln(df_{ij}) + {}_{ij}$ where the natural logarithm of absolute t-values, $ln(|t_{ij}|)$, is regressed on the natural logarithm of degrees of freedom, $\ln(df_{ij})$. If the slope coefficient, α_1 in MST is greater than zero then there exists a 'genuine' empirical effect. The intuition behind this is that for the existence of a 'genuine' relationship between milexp-growth, the magnitude of standardized test statistics, that is t-value in milexp-growth studies, varies positively with degrees of freedom, see Stanley (2001) and Stanley (2005). The results of the test are not reported here but they support the proposition of 'genuine' empirical effect.

Table 2
Results of fixed and random effects meta analysis.

Model		Fixed effects	Random effects
Number of studies		169	169
Mean effect size			
	Point estimate	0.056	0.066
	Lower limit (95% interval)	0.043	0.037
	Upper limit (95% interval)	0.069	0.095
Test of null			
	Z-value	8.35	4.46
	P-value	0.00	0.00
Heterogeneity test			
	Q-value	593.135	
	df (Q)	168	

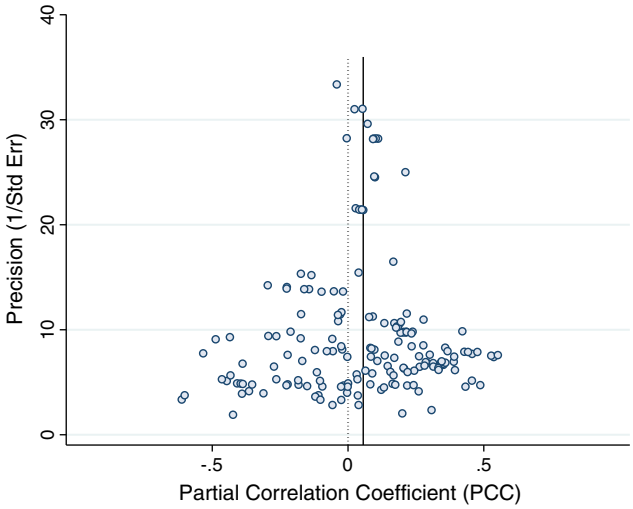

Fig. 1. Publication bias: funnel plot of PCC versus inverse standard error of PCC.

4.2. Multivariate meta-regression analysis

As noted in the preceding section, the assumption of homogenous ES is invalid for the milexp-growth studies. In other words, in the milexp-growth studies, the differences among ES are not due to sampling error alone. The aim of this section is to identify the factors that cause heterogeneity among ES. Due to heterogeneity, the results of the simple FAT test in the previous section may not be valid, Doucouliagos and Stanley (2008) and Stanley (2008). To model ES heterogeneity, a multivariate meta-regression analysis (henceforth, MRA) is used, this is also known as moderator variables analysis. The multivariate MRA also allows to further explore publication selection bias. The moderator variables refer to systematic variations in the original studies. These may emerge from the use of different theoretical reasoning, methodological issues or some other characteristics of empirical studies. For instance, in the milexp-growth literature, demand side theories expect a negative effect (Hypothesis 1) (via trade-off within other government expenditure) and supply side a positive one (Hypothesis 3) (via technological spillover and positive externalities). Keynesian demand and supply models, for instance, expect a negative association of milexp on growth through crowding out of other beneficial government expenditure; investment, health and education. However, given that milexp is one of the components of government expenditure and an increase in government expenditure through milexp can stimulate the economy via Keynesian multiplier. In effect, there is also an expectation of a positive effect through Keynesian multiplier effects.[22]

Furthermore, a moderator analysis allows for closer examination of individual studies. In what follows each study is reviewed and the main characteristics of individual studies are coded, which are then compared across all the studies. This provides variables that can be used as the moderating variables in the MRA. In other words, moderator variables allow us to segregate the role of other

[22] As mentioned in the narrative review, the neo-classical Feder–Ram supply-side models expect a positive effect or no economic effect.

Table 3
Funnel asymmetry test (FAT) & precision effect test (PET). Dependent variable: *t*-value.

Variable	Weighted least squares	Cluster data analysis	Hierarchical linear model
	(1)	(2)	(3)
$1/se_i$	0.049	0.049	0.099
	(2.12)*	(1.71)	(3.48)**
Constant	0.112	0.112	−0.476
	(0.43)	(0.31)	(1.10)
No. of estimates	169	169	169
No. of studies		32	32
R^2	0.03	0.03	

Note: Absolute value of t statistics in parentheses; * significant at 5%; ** significant at 1%.

control variables and theoretical–methodological issues on the study-to-study findings. Stanley and Jarrell (1989) suggest the following specification to model heterogeneity:

$$t_{ij} = \beta_0 + \beta_1 \frac{1}{SE_{ij}} + \sum_k^K \frac{\gamma_k Z_{k.ij}}{SE_{ij}} + e_{ij} \tag{2}$$

where t_{ij} is the *t*-value of the estimated partial correlation for estimate *i*, study *j*, *Z* denotes the individual study characteristics, and γ_k are the coefficients to be estimated, where each coefficient provides the estimated effect of a particular characteristic on the correlation coefficient.

Eq. (2) is a WLS MRA model of Eq. (1), see Doucouliagos and Stanley (2009). In general, OLS is used to estimate Eq. (2), which is a consistent estimator if the estimated milexp-growth effects of primary studies used to construct the sample are independent from one to another. However, primary studies, often, provide more than one estimate, in which case the independency among estimates can be questioned (de Dominicis et al., 2008). Some meta-analytic reviews use one estimate from each study to deal with this issue and subsequently use OLS, (Doucouliagos and Ulubasoglu, 2006). Instead of undertaking such selection procedure, Eq. (2) is estimated using a hierarchical model (also known as a multi-level model) to account for data dependency. It is, therefore, possible to use all the available information from each study and the equation to be estimated is:

$$t_{ij} = \beta_0 + \beta_1 \frac{1}{SE_{ij}} + \sum_k^K \frac{\gamma_k Z_{k.ij}}{SE_{ij}} + u_j + e_{ij} \qquad i = 1, 2, \cdots, L \tag{3}$$

where t_{ij} is the *i*th *t*-value derived from the *j*th study and *k* is the number of regressors. u_j is the study level error term, that allows each study to have varying intercept. This model is known as a random intercept or mixed-effects model. The error terms are assumed to have a normal distribution; $e_{ij} \sim N(0, \sigma_e^2)$ and $u_j \sim N(0, \sigma_u^2)$. The error terms are independent, thus the total variance is given by; $\sigma_u^2 + \sigma_e^2$.[23] Using the variance of the error terms, the within-study correlation is: $\rho = \sigma_u^2/(\sigma_u^2 + \sigma_e^2)$.

The mixed-effect model allows us to account for between study heterogeneity. It assumes that the variability among effect size (PCC) is larger than one would expect merely from sampling error. Hence, the model allows for both sampling error and the study-level random component. If the study-level error term is not significantly different from zero, then the model reduces to fixed-effects meta analysis, Thompson and Sharp (1999).

4.2.1. Results of multivariate MRA

Based on the discussion in Sections 2 and 3, there are identified three broad areas of methodological differences that may influence the investigation of milexp-growth effect: (a) econometric specifications, (b) underlying theoretical models, and (c) other control variables in the primary study. In total, there are identified twenty-one potential moderator variables.[24] The definition of individual variables that are used in meta-regressions is given in Table A.2. All the independent variables used are dummies with the exception of the natural logarithm of the number of observations and number of years in the primary study.

Table 4 reports the results of the weighted least squares (WLS), clustered data analysis (CDA) and hierarchical linear model (HLM). The dependent variable is *t*-values. There are two specifications, general and specific. Odd columns provide results for more general specifications and even columns general-to-specific. The specific model is obtained by excluding one by one the insignificant variables in the general model using CDA. At each step, the Ramsey RESET test is used to test the effect of omitted variable. Although, in column (4), both the 'Single Country" and SEM are not significant at conventional level, they affect the model diagnostics. Therefore, these variables are not excluded. The discussion here is mainly based on HLM results. WLS and CDA are given only for comparison purpose.

[23] A detailed description of multilevel techniques can be found in Goldstein (1995) and Hox (2002).
[24] Note that not all these identified control variables are used in the regressions.

Table 4
Meta-regression analysis results of milexp-growth studies. Dependent variable: *t*-value (t_{ij}).

Variable	Weighted least squares		Clustered data analysis		Hierarchical linear model	
	General (1)	Specific (2)	General (3)	Specific (4)	General (5)	Specific (6)
$1/SE_{PCC}$	−0.074	0.086***	−0.074	0.086***	−0.086	0.097**
	(−0.33)	(4.43)	(−0.32)	(5.06)	(−0.31)	(2.94)
Number of observations and years in the primary study						
lnobs	0.036		0.036		0.039	
	(1.17)		(1.01)		(0.95)	
lnyear	0.096*		0.096*		0.119	
	(2.45)		(2.35)		(1.56)	
Econometric specifications and underlying theoretical models used in the primary study						
D50	0.188**	0.221***	0.188***	0.221**	0.258**	0.344***
	(2.99)	(3.93)	(4.10)	(3.47)	(2.72)	(4.28)
D60	−0.0931		−0.0931		−0.116	
	(−1.94)		(−1.54)		(−1.64)	
D80	−0.131		−0.131		−0.165	
	(−1.26)		(−1.16)		(−1.17)	
D90	−0.176***	−0.194***	−0.176***	−0.194***	−0.133*	−0.130**
	(−4.36)	(−5.61)	(−4.07)	(−3.89)	(−2.29)	(−2.79)
ACDA	−0.0579		−0.0579		−0.0743	
	(−1.35)		(−1.23)		(−1.66)	
Developed	0.198***	0.202***	0.198**	0.202**	0.169*	0.169**
Countries	(3.45)	(4.47)	(2.86)	(3.65)	(2.55)	(3.28)
Africa	0.0227		0.0227		0.0131	
	(0.52)		(0.51)		(0.22)	
Panel	−0.143		−0.143		−0.184*	
	(−1.83)		(−1.62)		(−2.47)	
Single	−0.208	−0.154*	−0.208	−0.154	−0.276	−0.201
Country	(−1.94)	(−2.00)	(−1.63)	(−1.45)	(−1.72)	(−1.33)
SEM	0.032	−0.043	0.032	−0.043	0.090	−0.030
	(0.74)	(−1.50)	(0.54)	(−1.51)	(1.61)	(−0.79)
Growth theory	−0.017		−0.017		−0.067	
	(−0.29)		(−0.24)		(−0.98)	
Other control variables used in the primary study						
Aid	−0.098		−0.098		−0.127	
	(−1.95)		(−1.96)		(−1.70)	
Dem	0.160**		0.160**		0.148*	
	(3.23)		(3.08)		(2.27)	
FDI	0.149*		0.149**		0.111	
	(2.46)		(2.99)		(1.11)	
NME	0.151**	0.073*	0.151**	0.073	0.114	0.090
	(2.92)	(2.10)	(2.78)	(1.72)	(1.64)	(1.28)
Edu	−0.052		−0.052		0.0004	
	(−1.03)		(−0.71)		(0.01)	
Non-linear	0.109**	0.091**	0.109	0.091*	0.103**	0.076*
Functional forms	(2.79)	(2.91)	(2.04)	(2.52)	(2.66)	(2.11)
Constant	−0.535	−0.476	−0.535	−0.476	−0.609	−0.654
	(−0.83)	(−1.83)	(−0.85)	(−1.43)	(−0.72)	(−1.73)
ρ					0.40	0.39
N	169	169	169	169	169	169
S			32	32	32	32
R^2	0.53	0.36	0.53	0.36		

Note: Each moderator variable is divided by the associated standard errors, SE_{pccij}.
N and S are the total number of estimates and studies respectively.
t statistics in parentheses.
* $p<0.05$, ** $p<0.01$, *** $p<0.001$.

First, the effects of the number of observations and number of years are considered. Most of the previous studies conduct a cross-country analysis and cover very short-time periods, which may be due to the fact that in the long-run there are much less variations in the milexp series. Therefore in addition to the sample size variable, the number of years is used to see if this will cause any significant effect among reported results of the primary studies. In the general model, the sample-size variable is statistically insignificant and the number-of-years variable is negative and significant for WLS and CDA. The studies that have analysed the growth-milexp relationship using longer time-period report a weaker relationship. One explanation to this finding is that in the

long-run, there is very little variation in average milexp. For example, the world average milexp was in the range of 3.0–4.5% of GDP over the period of 1960–2002.[25] The volatility in milexp is mostly observed in countries that are in internal/external conflict but, in general, a period of high spending follows a low spending.

Second, the effect of econometric specifications is considered. The following control variables are used to capture this effect: time periods, data sources, sample of countries, and estimation types. The effect of time periods are captured using five dummies; 1950s (D50), 1960s (D60), 1970s (D70), 1980s (D80), and 1990s (D90). 1970s is used as the base. The period dummies are also important to attest whether the estimated milexp-growth effect is time invariant. Both specifications in columns (5) and (6) show that the time periods contribute significantly to the variations in the reported findings. In the general specification, the dummies for 1960s 1980s and 1990s are negative, that means reducing the milexp-growth effect, whereas the period of 1950s has a significant positive effect. For the year dummies 1990s and 1950s, the same conclusions emerge from the specific model reported in column (6). These suggest that the milexp-growth effect changes over time.

The milexp data used in the empirical studies are mainly from two sources; the United States Arms Control and Disarmament Agency (ACDA) and the Stockholm International Peace Research Institute (SIPRI). Although, both ACDA and SIPRI use the NATO definition of milexp, they differ somewhat. ACDA is included in the MRA analysis with SIPRI as the base. In the general model, ACDA is negative and significant at 10%. However, the use of different data sources does not affect the reported milexp-growth effect.[26]

Also, a set of dummy variables are defined to capture the effect of the type of data, namely 'Panel' for the panel data and 'Single Country' for the time series data. Using the cross-section data as the base, column (5) shows that both dummies are negative and significant at 5% and 10% respectively. Also, the dummy variables 'Developed Countries' and 'Africa' are used to assess the effect of a specific group of countries. The variable 'Developed Countries' is positive and significant in both specifications. The dummy variable for 'Africa' is negative but statistically insignificant. In effect, given the importance of hypothesis 2, more research is necessary to investigate homogenous samples at regional and country level.

Third, three dummy variables are considered for theoretical models. 'Growth theory' is a dummy variable taking the value 1 if the theoretical model is either the endogenous or exogenous (neo-classical) growth model. The effect of conducting a Keynesian demand–supply model on the correlation coefficient is captured by another dummy variable, simultaneous-equation model (SEM). 'Growth other' is a dummy variable taking value 1 for all other models.[27] Using 'Other model' dummy variable as the base, the dummy variables for growth theory and SEM are included in the MRA. Both dummies are statistically insignificant in all specifications. However, two points should be noted. First, it was not possible to include the net effect from individual studies that apply the SEM. Hence, the direct effect is only considered. Second, it would have been ideal to sub-divide competing growth theories. Since conceptually, the aggregated/disaggregated government expenditure (including milexp) can hamper/boost economic growth. In the neo-classical growth theory, government expenditure does not influence the long-run economic growth, whereas the endogenous growth models show that fiscal policy has important implications on the long-run growth, Barro (1990).

Finally, six dummy variables are considered to account for the other control variables used in the primary studies. These are aid (Aid), democracy (Dem), foreign direct investment (FDI), neighbouring countries' milexp (NME), education (Edu) and non-linear functional forms (or interaction term). The dummy variables Edu, FDI and NME are statistically insignificant in both specifications for HLM. However, FDI and NME are significantly positive in WLS and CDA. In the general specification (column 5) the dummy variables for non-linear functional form is positive and statistically significant at 5% level. The non-linear effect of milexp on growth is discussed more commonly in recent empirical studies. This finding appears to complement that discussion. Thus, the studies that account for a non-linear relationship reports a stronger link between growth and milexp.

The net effect is now determined by all the coefficients of moderator variables and 1/*SE*. However, using column (6), if all the moderator variables are zero, then the estimated effect size is 0.097, which is genuine empirical milexp-growth effect. Overall, the characteristics of studies systematically affect the estimated milexp-growth relationship to vary, in particular the time periods, the sample and the functional forms among others. Also, the preferred model (column 6) indicates a small publication selection bias.

5. Discussion

In the literature, there are several channels identified that may simultaneously produce the positive and negative impacts of milexp on economic growth. For certain countries, in particular some LDCs, it has been widely discussed that the negative effects might be more pronounced than positive effects. Although, there has been a general understanding of mixed results in individual empirical studies, there is no consensus as to the direction of the net empirical effect of milexp-growth relationship, which has been difficult to quantify. Therefore, in this systematic review, the costs and benefits of milexp to the economy are formulated into four hypotheses: Military expenditure reduces economic growth, (H1); military expenditure is detrimental to economic growth in LDCs,

[25] Authors' own estimate.

[26] It should be noted that two of the international institutions, namely the NATO and the UN, provide a definition of milexp that differ in some levels. The NATO definition of milexp includes expenditure on police forces and the UN peace forces but these are excluded from the UN definition of milexp. On the contrary, the UN includes military aid but not the NATO (Brzoska, 1995). None of the studies included in this meta-analysis uses data from the UN.

[27] See Table A.1 in which the studies that apply the Keynesian demand–supply model are classified as *approach* ‡ the studies that apply the formal economic growth models are classified as *approach§* and other approaches are classified as *approach O*.

(H2); the effect of military expenditure on economic growth is positive, (H3); the effect of military expenditure on economic growth is non-linear, (H4). On the basis of our meta-analysis of growth-milexp literature it is clear that (H1) and (H2) are rejected but (H3) and (H4) are supported. The findings reported in columns 4 and 6 of Table 4 show that the coefficient of 1/*SE* is positive and statistically significant, 0.086 and 0.097 respectively. Although, the overall effect is positive, according to Cohen's standards, it is practically negligible. Moreover, the upper boundary of the net effect size is very close to 0.10, arguably, it could be concluded that the effect is not trivial. In addition, Doucouliagos (2011) argues that applying Cohen's standards to partial correlations understate the economic significance of the empirical effect. Therefore, a new set of guidelines is proposed, in which an effect size greater than 0.07 is considered as 'medium', see Doucouliagos (2011).

Using the coefficient of the dummy for Africa, our meta-analysis shows that the effect of milexp on growth in LDCs is statistically and practically insignificant, column 5 in Table 4. Some LDCs have been through considerable internal conflicts with very poor economic performance. The high risk of turmoil and conflict may potentially increase demand for milexp. Collier and Hoeffler (2002) look at the possible effect of external and internal conflicts on milexp. Their findings show that internal conflict is one of the important factors for high level of milexp. They also find that milexp is not effective in reducing the risk of internal conflict. Besides, the level of income and economic growth have a negative impact on civil war. Therefore, considering the complexities of the conflict-milexp-growth nexus in LDCs and also the difficulty to measure a thorough conflict variable that is not only proxied by war or instability dummies, it is not unanticipated that (H2) is rejected.

(H3) is supported for the developed countries. The coefficient on developed countries is consistently positive and statistically and practically significant across the specifications in Table 4. In particular, in our preferred specifications, the coefficient is 0.202 and 0.169, in columns 4 and 6 respectively. This implies that the primary study that includes developed countries in the analysis reports a much larger positive milexp-growth effect on average. One possible explanation is that most developed countries may still be at a low level of milexp relative to LDCs, in which case the benefits outweigh the opportunity cost of milexp. Also, it should be noted that many developed countries are involved in military research and development (R&D), which is arguably considered to have positive externalities on civilian sector. However, the effect of technological spin-off from military R&D to the civilian sector is also very controversial, see Smith (2008).

(H4) suggests that there can be circumstances where positive effects on economic growth arising from an increment in milexp could eventually be dominated by the opportunity cost of diverting resources away from the civilian sector of the economy. Potentially turning the net effect of milexp on growth from positive to negative. Our meta-analysis shows that the studies that formulate a non-linear milexp-growth relationship produce a strong positive effect. Although the magnitude of the non-linear variable is very small, 0.091 and 0.076, columns (4) and (6) in Table 4 respectively, (H4) is confirmed.

Finally, our bivariate MRA results are also in agreement with the multivariate analysis. Table 3 shows that the selection bias corrected averages are 0.049, for WLS and CDA, and 0.099, for HLM, in what follows, the net effect of milexp on economic growth is positive but practically negligible.

6. Summary

This paper provides a meta-analytic review of military expenditure (milexp) and economic growth literature. It consists of 32 empirical studies and 169 estimates. The following conclusions emerge from the meta-regression analysis (MRA). First, our bivariate and multivariate MRA show that the combined effect of milexp-growth studies is estimated to be positive and ranges between 0.056 and 0.099. Second, the hypothesis of a negative milexp-growth relationship is not supported for both LDCs and in general, while a positive effect of milexp on growth is supported for developed countries. Third, the hypothesis of a non-linear milexp-growth relationship is also confirmed. Fourth, the main methodological differences in the primary studies that contribute to the variations in the findings of the existing literature are the sample, time-periods, and functional forms. To some extent, the findings of this complementary approach support formally the general conclusions of the earlier qualitative surveys, see Dunne (1996) and Ram (1995).

We have identified a number of shortcomings in the literature, which may help defining possible routes for future search. First, only a few primary studies of the milexp-growth include more recent time periods. Most of the primary studies use the time period of data covering 1960s, 1970s and 1980s. Therefore, the future research should focus on extending the time periods to cover 1990s and onwards. After all, with the end of the Cold War in 1990s, the perceived threat is reduced across nations and a knock-on effect on milexp seems plausible. Second, although the argument of a negative effect of milexp on growth is stronger for the poorer LDCs, it was not possible to run a separate MRA due to the small sample for this category. Third, as suggested in many recent studies and also in our meta-analysis, the non-linear functional forms appear to be an important issue. Clearly, there is a scope to continue exploring this in future research. Finally, in this MRA, the milexp-growth *effect size* is the partial correlation coefficient between economic growth and the *level* of milexp. Therefore, the studies that analyse the *growth* of milexp and economic growth are left out in this meta-analysis, i.e. the Granger causality and Feder–Ram based studies. Hence, the conclusions drawn from this MRA should be carefully interpreted and a separate meta analysis of Feder–Ram based studies could offer further insights.

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Appendix A

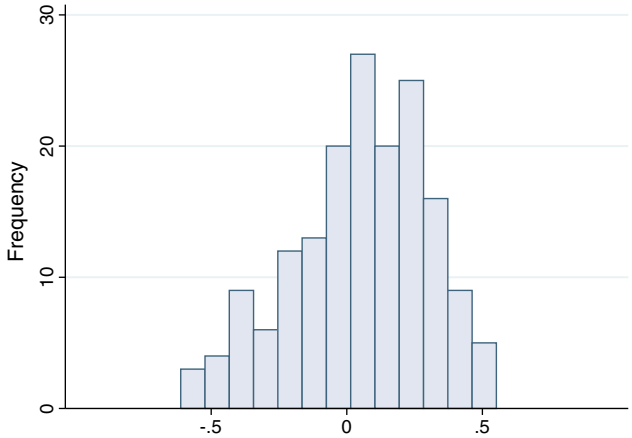

Fig. A.1. Distribution of partial correlation coefficient.

Table A.1
Military expenditure–growth studies included in the meta analysis.

Study	Approach	Data type	Data period
Benoit (1978)	*O*	Cross section (linear)	1950–1965
Deger and Smith (1983)	‡	Cross section (linear)	1965–1973
Lim (1983)	§	Cross section (linear)	1950–1973
Deger and Sen (1983)	‡	Cross section (linear)	1965–1973
Faini et al. (1984)	*O*	Time series (linear)	1950–1972
Cappelen et al. (1984)	*O*	Panel (linear)	1960–1980
Landau (1986)	*O*	Panel (linear)	1960–1980
Biswas and Ram (1986)	*O*	Cross section (linear)	1960–1977
Deger (1986)	‡	Cross section (linear)	1965–1973
Lebovic and Ishaq (1987)	‡	Panel (linear)	1973–1982
Chan (1988)	*O*	Time series (linear)	1961–1985
Grobar and Porter (1989)	*O*	Cross section (linear)	1950–1965
Gyimah-Brempong (1989)	‡	Panel (linear)	1973–1983
Looney (1989)	*O*	Cross section (linear)	1970–1982
Landau (1993)	*O*	Panel (nonlinear)	1969–1989
Dunne and Mohammed (1995)	‡	Panel (linear)	1967–1985
Lipow and Antinori (1995)	*O*	Cross section (linear)	1974–1989
Blomberg (1996)	§	Panel (linear)	1967–1982
Knight et al. (1996)	§	Cross section & Panel (linear)	1971–1985
Landau (1996)	*O*	Panel (nonlinear)	1950–1990
Brumm (1997)	§	Cross section (linear)	1974–1989
Antonakis (1997)	*O*	Time series (linear)	1960–1990
Heo and DeRouen (1998)	*O*	Time series(linear)	1961–1990
DeRouen (2000)	*O*	Time series (linear)	1953–1992
Stroup and Heckelman (2001)	§	Panel (nonlinear)	1975–1989
Galvin (2003)	‡	Cross section (linear)	1999
Aizenman and Glick (2006)	§	Cross section (nonlinear)	1989–1998
Bose et al. (2007)	§	Panel (linear)	1970–1989
Kalaitzidakis and Tzouvelekas (2007)	§	Panel (nonlinear)	1980–1995
Kollias et al. (2007)	*O*	Panel (linear)	1961–2000
Yakovlev (2007)	§	Panel (linear–nonlinear)	1960–2000
Looney and McNab (2008)	*O*	Panel (linear)	1999–2003

Approach ‡ represents a model that is based on either the short-run Keynesian demand–supply model of Deger–Smith or SEM methods.
Approach § represents that the empirical specification is derived from either the long-run Solow or Barro type growth model.
Approach *O* represents the other models, which do not fall in to either category ‡ or §.

Table A.2
Meta-regression analysis variables.

Variables	Definition
No. observation in primary study (lnobs)	Natural logarithm of number of observation in the primary study.
No. years in primary study (lnyear)	Natural logarithm of number of years in the primary study.
Dummy 1950s (D50)	Takes value 1 if the primary study includes data from 1950s, otherwise 0.
Dummy 1960s (D60)	Takes value 1 if the primary study includes data from 1960s, otherwise 0.
Dummy 1980s (D80)	Takes value 1 if the primary study includes data from 1980s, otherwise 0.
Dummy 1990s (D90)	Takes value 1 if the primary study includes data from 1990s, otherwise 0.
ACDA	Takes value 1 if the data used in the primary study comes from ACDA, otherwise 0.
Developed countries	Takes value 1 if the primary study data is from developed countries, otherwise 0.
LDCs	Takes value 1 if the primary study use data only from LDC's, otherwise 0.
Africa	Takes value 1 if only African countries data is used, otherwise 0
Single country study	Takes value 1 if a single time series analysis of a country is done, otherwise 0.
Simultaneous equations model (SEM)	Takes value 1 if the method of estimation is simultaneous equations approach, otherwise 0.
Panel	Takes value 1 if the method of estimation is panel data in the primary study, otherwise 0.
Growth theory	Takes value 1 if the model is based on economic growth model, otherwise 0.
Aid	Takes value 1 if the primary study control for AID, otherwise 0.
Non-linear functional forms	Takes value 1 if the primary study control for non-linear or interaction terms, otherwise 0.
Democracy (Dem)	Takes value 1 if the primary study control for democracy, otherwise 0.
Education (Edu)	Takes value 1 if the primary study control for education, otherwise 0.
Instability or war	Takes value 1 if the primary study control for instability or war, otherwise 0.
Foreign direct investment (FDI)	Takes value 1 if the primary study control for FDI, otherwise 0.
Neighbouring countries' milexp (NME)	Takes value 1 if the primary study control for NME, otherwise 0.

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